WorldMark: A Unified Benchmark Suite for Interactive Video World Models

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Research Questions

- How can we fairly compare interactive video world models that use heterogeneous control interfaces and private evaluation benchmarks?
- What are the relative strengths and weaknesses of current interactive I2V models across visual quality, control alignment, and world consistency?
- How do interactive video models perform differently in first-person vs third-person viewpoints and across photorealistic vs stylized scenes?

Core Idea

Interactive video world models like Genie, YUME, and HY-World each use different control formats (captions, pose parameters, gamepad controls) and evaluate on private benchmarks, making cross-model comparison impossible. WorldMark solves this by creating a unified action-mapping layer that translates standardized WASD+L/R commands into each model's native format, enabling apples-to-apples evaluation. The key insight is that standardized test conditions (same scenes, same actions) are more important than sophisticated metrics - even simple metrics reveal meaningful differences when models are evaluated fairly on identical inputs.

Key Formula

Translation Error

$$e_t = \|t_{gt} - st\|_2$$

t_gt and t are ground-truth and estimated camera positions in R^3, s is least-squares scale factor. Measures spatial consistency between generated video and conditioning camera trajectory using scale-invariant Euclidean distance.

Reprojection Error

$$e_{reproj} = \frac{1}{|V|} \sum_{(i,j) \in V} \|p_{ij}^* - \Pi(P_{ij})\|_2$$

V is set of co-visible pixel pairs, p*_ij is observed 2D coordinate, P_ij is reconstructed 3D point, Π denotes camera projection. Assesses 3D spatial coherence via DROID-SLAM reconstruction - higher error indicates geometric inconsistency and scene structure collapse.

Key Figure

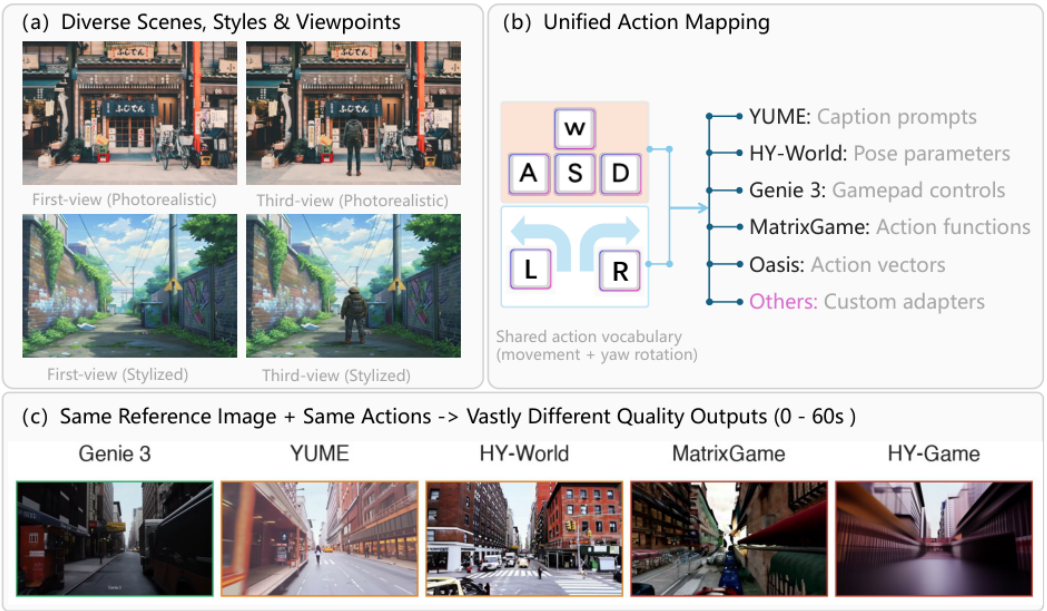

Figure 1 Overview of `WorldMark`. Current interactive world models each define their own control interfaces, scenes, and evaluation protocols, making cross-model comparison impossible. `WorldMark` resolves this fragmentation through three pillars: **(a)** A curated test suite of 500 cases spanning first- and third-person views with both photorealistic and stylized reference scenes, ensuring broad coverage of real-world evaluation conditions. **(b)** A unified action-mapping layer that translates a shared action vocabulary—WASD for movement and L/R for yaw rotation—into each model's native control format (caption prompts, pose parameters, gamepad signals, action vectors, etc.), so that all models receive semantically identical instructions. **(c)** Under these standard conditions, feeding the *same* reference image and *same* action sequence to different models reveals striking quality divergence that proprietary benchmarks obscure.

Figure 1: Shows the core problem and solution - current models use different control interfaces and private benchmarks, while WorldMark enables fair comparison by translating shared WASD actions into native formats, revealing striking quality differences.

Limitations

- Limited to 6 specific models and may require new action-mapping adapters for future models with novel control interfaces
- VLM-based scene filtering and evaluation metrics may introduce bias or inconsistency compared to human judgment
- Test suite of 50 reference images may not capture full diversity of real-world scenarios and edge cases
- Third-person viewpoint evaluation restricted to only 3 models that natively support this perspective